

Constraints on the Moon, and the flaws in this simulation

What the lunar surface actually imposes, what viewer.html models today, and what is still wrong. Written against the code in Hunch/nasa-llaso-cad/viewer.html as of 23 September 2026.

Scope. This is the driving simulation: six wheels, lidar occupancy map, onboard A* and pure pursuit, and hunch-rover-nav on the Mac Mini at 100.92.134.100:11434. It is not a thermal, power, or structural analysis of flight hardware.

1. Hard constraints of the lunar surface

These numbers are the environment. A rover that ignores one of them is simulating a different world. Sources are the NASA Moon Fact Sheet, the Lunar Sourcebook, Apollo soil-mechanics experiment S-200, and the Bekker parameters used in lunar terramechanics.

1.1 Gravity, orbit, and the day

Quantity	Moon	Earth	Ratio
Surface gravity	1.62 m/s ²	9.80 m/s ²	0.165
Escape speed	2.38 km/s	11.2 km/s	0.213
Mass	7.35×10 ²² kg	5.97×10 ²⁴ kg	0.0123
Equatorial radius	1738 km	6378 km	0.273
Mean density	3344 kg/m ³	5514 kg/m ³	0.61
Sidereal month	27.32 days	—	tidally locked
Solar day	29.53 days	24 h	about 29.5×
Mean distance	384,400 km	—	light time ~1.3 s
Obliquity to orbit	6.68°	23.4°	weak seasons
Bond albedo	0.11	0.31	dark surface
Solar irradiance	1361 W/m ²	1361 W/m ² at Earth orbit	1.0, no air loss

A 1 m drop takes about 1.11 s and ends at 1.8 m/s. The same drop on Earth takes 0.45 s and ends at 4.4 m/s. With a traction coefficient of 0.55, stopping from 1 m/s takes about 0.56 m on the Moon and 0.09 m on Earth. Slides are longer because weight, and therefore friction force, is smaller.

1.2 Vacuum, plasma, and heat

- The entire lunar atmosphere is about 25,000 kg. Nighttime surface pressure is about 3×10^{-14} bar, near 2×10^8 molecules per cm³. There is no aerodynamic drag and no convective cooling.
- Equatorial ground runs from about 95 K to 390 K (−178 °C to +117 °C) over one solar day. The daily heat wave dies within about a meter of regolith because conductivity is very low.
- Heat leaves a rover only by radiation and by conduction into the soil or into a radiator. A battery, motor, and avionics box that rely on air will overheat in the day and freeze in the night.

- Sunlit regolith charges positive. Shadowed regolith charges negative. Apollo dust adhered at about 10^{-4} dyn/cm² on paint and $2\text{--}3 \times 10^3$ dyn/cm² on metal. Grains follow ballistic arcs. They do not drift in a breeze.
- There is no global magnetic field to deflect solar wind. The surface sees solar energetic particles and galactic cosmic rays directly.

1.3 Regolith and what a wheel may do

Property	Constraint	Driving consequence
Bulk density	About 0.7–1.5 g/cm ³ in the top layer, above 1.5 g/cm ³ by 10–20 cm	The surface is loose. Strength rises quickly with depth.
Cohesion	Typically 0.1–1 kPa. Engineering models often use 170 Pa.	Treads and footprints hold a sharp shape.
Friction angle	30–50°. Canonical terramechanics value is about 33°.	Loose grains lock. Steeper than dry beach sand.
Angle of repose	Near the friction angle, roughly 30–40°.	Fresh crater rims sit near this. Steeper faces slump.
Bekker set	$n = 1$, $k_c = 1400$ N/m ² , $k_\phi \approx 8.2 \times 10^{-4}$ N/m ³	Sinkage follows wheel pressure, width, and slip.
LRV tracks	Mean sinkage 1.25 cm, up to about 5 cm on fresh rims	Flotation was good. The mesh wheel sheared the surface.
LRV slopes	Apollo 15 to 12°, 16 to 18°, 17 to 20°	Twenty degrees was the flown limit, not the theoretical one.
LRV slip	About 2–5% on level ground. Navigation assumed 2.3%.	Odometry that ignores slip walks off the true pose.
Climb limit	$\tan \theta = \mu$. At $\mu = 0.55$ that is 29°. At $\phi = 33^\circ$, $\tan \phi = 0.65$.	A wheel that exceeds this shears and digs instead of climbing.

Downslope acceleration is $g \sin \theta$. At 20° that is 0.55 m/s² on the Moon and 3.35 m/s² on Earth. The rover is light, so the pull is gentle, but it does not go away, and the same lightness means less force is available to brake it.

1.4 Power, light, and operations

- A solar day is 14.8 Earth days of sun and 14.8 of night. A battery sized for hours of driving does not survive the night without a heater and a shutdown plan.
- Solar flux is 1361 W/m² only while the panel is clean and pointed. Dust adhesion cuts that. There is no wind to clear a panel.
- One-way light time to Earth is about 1.3 seconds. Teleoperation that assumes zero delay will overshoot.
- Line of sight to Earth is blocked by terrain and by the far side. A rover in a crater can lose the link without any hardware failure.
- Wheel sinkage, slip, and slope all change the energy per meter. A flat-ground watt-hour budget underestimates a crater rim.

2. What the simulation already gets right

These are in viewer.html now. They should stay.

Item	In the sim	Lunar value
Gravity	$G = 1.62$ m/s ²	1.62 m/s ²
Slope force	$g \sin \theta = -\sin(\text{pitch}) \cdot 1.62$, applied after the traction cap	$g \sin \theta$, full strength
Traction cap	$\mu g \cos \theta$ with $\mu = 0.55$. Extra drive force is slip.	Order of a mesh wheel, not a sticky tire
Rolling loss	0.05 g opposing velocity	Same order as LRV rolling resistance

Item	In the sim	Lunar value
Sinkage	1.25 cm baseline, grows with slip and pitch , capped at 5 cm	Apollo mean 1.25 cm, rims about 5 cm
Wheels	Six separate tires, hubs, chevrons, and struts	Replaces the continuous side bar
Speed	0.94 m/s from 100 rpm and a 9 cm rolling radius	Hardware choice, slower than the LRV
Sensing	Spinning lidar builds the occupancy map. The planner does not read the rock list or the crater catalog.	A real rover does not get a god map

3. Flaws, in the order they change the drive

Each flaw is something the current code still does. Checked against viewer.html, a summit drive, a flat-ground run, and a 90-second wander on the 144-rock field.

3.1 The motor is not a motor

activeMotor.torque is 12, then the code uses $\text{torque} \times 2$ as an acceleration in m/s^2 , capped at 24 m/s^2 . Torque is not acceleration. There is no vehicle mass, so a $12 \text{ N}\cdot\text{m}$ motor and a chassis of unknown kilograms are the same number. The traction cap then throws almost all of that away: any real throttle command exceeds $\mu g \cos \theta$ (about 0.89 m/s^2 on flat ground), so the slip flag is true whenever the rover is asked to speed up or slow down hard. The slip light is measuring the unit error, not the soil.

The rolling radius is also split. Visual tire radius and contact height use 0.10 m. Speed, RPM, and wheel spin use 0.09 m. At the same ground speed the drawn wheel and the odometry disagree by about 11 percent.

3.2 Soil is still a heightfield with a cosmetic sink

- Sinkage is a formula of slip and slope, then a vertical offset. It does not come from wheel pressure, width, or the Bekker k_c / k_ϕ set. A heavier load would not sink more, because mass is not in the model.
- Cohesion is absent. Footprints and chevrons would not hold, and a wheel would not have the small cohesive contribution to thrust that Apollo soil had.
- There is no bulldozing term. On a 3 cm rut the real resistance is the soil pushed ahead of the tire. Here the rut does not push back.
- Strength does not increase with depth. The top 15 cm on the Moon is the weak layer. Under it, excavation force rises. The sim's 5 cm cap is a picture of that, not the mechanism.
- Internal friction of $30\text{--}50^\circ$ is not the failure surface. $\mu = 0.55$ implies a slide near 29° . That is plausible for traction, and it is still a single number for maria, rims, and crater fill.
- Driven slopes in the mesh reach 0.85 rad , about 49° , which is past the angle of repose. Apollo stopped near 20° . The rover can be asked to climb a face that would have slumped.

3.3 Dust, vacuum, and heat are the wrong physics

updateDust gives grains an upward velocity and then accelerates them downward at 0.3 m/s^2 . Lunar gravity is 1.62 m/s^2 , and there is no air to slow a grain. The particles hop and die in 0.7–1.1 s. They do not follow a vacuum ballistic arc, they do not stick to the solar deck, and they do not charge.

- No thermal model. Equatorial hardware would cycle between about 95 K and 390 K. Motors, battery, and the camera are the same temperature on the night side as at noon.
- No radiative balance. Solar input is not 1361 W/m^2 on the panel, and the panel is not derated by dust.
- The battery is 240 Wh (20 Ah at 12 V) with a flat 55 W draw while moving and 15 W while sitting, and +100 W at the dock. Night survival, heater power, and voltage sag in the cold are absent. Draw does not rise on a slope or in

slip.

- Dock contact is a 2.9 m sphere around the pad, using 3D distance, so a rover on a rise can count as docked, and a rover beside the pad on a slope can miss.
- One-way light time and terrain occultation are absent. The operator click and the model reply are limited only by the Mini's inference time.

3.4 Navigation still fails in ways a slope and a lie will cause

- Lidar beams pitch downward (-0.05 , -0.12 , -0.21 rad). On flat ground the lowest beam hits soil a few meters out. That range is easy to confuse with an obstacle if the obstacle test is too strict. A miss used to be painted as a wall; that paint is now a mild slope, but the beams still do not separate "ground" from "rock" the way a real return's normal and range residual would.
- Occupancy near the rover fades by 0.992 per sweep when the log-odds are under 3.2. A rock seen once and then left beside the path can disappear before the planner needed it. A rock kept in view stays.
- The stuck watchdog marks a cell only on flat ground. On a slope it replans without a mark. In a 90 s wander that still fired 5 times. The rover kept moving, but the route is being thrown away often.
- Goal distance for success is 2.0 m horizontal. The follower starts stopping inside 1.2 m. On a summit those two thresholds, plus a slow creep, are how the rover used to sit short of the flag. Horizontal distance is fixed. The creep on a steep face is not.
- Wander goals are uniform random in a 100 m box. They can land inside a rock, on a 49° face, or in a crater bowl. The planner only knows what the lidar has seen, so the first plan is often a straight line into unseen terrain.
- hunch-rover-nav on the Mini is a text bias from three front ranges. llama3.2-vision does not load: that Ollama build rejects the mllama architecture, which is why the badge showed THINKING then ERR. The camera is not in the navigation loop.
- The model is asked every 1.8 s and can add a left/right bias on top of A*. That bias can fight a path that was already clear.
- Odometry assumes the commanded velocity, with no 2–5% slip term. After a long traverse the pose the planner uses and the pose a wheel encoder would report are the same, which is the failure Apollo's 2.3% slip bias was built to avoid.
- A 1 hour wander was started and aborted before a full hour of statistics existed. The 90 s sample covered 50 m, reached 1 goal, closest rock 1.1 m, 0 crashes, 5 stuck replans. That is not an endurance result.

3.5 Graphics and the page

- Earthshine and a blue Earth are in the sky, which is fair, but the surface is lit like a game level: exponential fog, a 2048 shadow map, and no 14-day terminator crawl.
- The six wheels roll, but the suspension stroke is still a ± 4.5 cm cosmetic hug of the heightfield. It is not a rocker-bogie with a differential.
- Safari was the buggy client. Scissor picture-in-picture, backdrop blur, logarithmic depth, and 100vw sizing were removed. That was not re-checked inside Safari itself after the later physics edits.
- index.html in the same folder is an older copy of the page. Opening it shows the previous rover and the previous physics.
- Client errors are appended to rover_actions.log with no rotation. The file is already past 19,000 lines, including stale trailPoints errors from an older build.

4. What to change, in order

Order	Change	Why it is first
1	Give the rover a mass. Turn 12 N·m into wheel force, then into acceleration. Stop treating torque as m/s ² .	Until this is done, slip and sinkage are flags, not soil.
2	One rolling radius. Use 0.10 m for speed, RPM, spin, and contact.	11% odometry error is already larger than Apollo's slip bias.
3	Sink from pressure: Bekker z with $n = 1$, $k_c = 1400$, $k_\phi \approx 8.2 \times 10^4$, plus a bulldozing term when z and slip rise.	Matches the 1.25 cm / 5 cm Apollo envelope for a reason.
4	Refuse slopes past about 20–25°, or let soil past the friction angle slump. Do not offer 49° faces as drivable.	That is the flown limit and the repose limit.
5	Dust under vacuum: ballistic arcs at 1.62 m/s ² , no air drag, grains that stick to the panel and cut solar power.	The current 0.3 m/s ² hop is neither Moon nor Earth.
6	Battery draw that grows with slope, slip, and sink, and a night case. 240 Wh at 55 W is about 4.4 h of driving and does not survive a lunar night.	Power is the constraint that ends a traverse.
7	Keep the lidar map, and add a 2–5% slip bias on the pose. Decay occupancy more slowly than one glance.	The planner is the right architecture. The memory and the odometry are short.
8	Leave llama3.2-vision off this Mini until Ollama can load mllama. Keep hunch-rover-nav as a bias, not as a second driver.	The vision call is a known hard error.

5. A short record of what was measured

Run	Result
Flat ground, throttle held	Speed settled near 0.89 m/s. Sink 1.3–1.4 cm. Pitch about 0–2°.
Summit of the large hill ($r = 10$ m) to a goal 1.4 radii downrange, before full $g \sin \theta$	Reached in 13.8 s. No crash.
Same summit after full lunar slope force and sinkage	Reached in 18.1 s. Peak pitch –24.6°. Peak sink 3.3 cm. No crash.
90 s random wander, 144 rocks	50 m, 1 goal, closest approach 1.1 m, 0 crashes, 5 stuck replans.
Vision model on the Mini	llama-server exit 1, unknown architecture mllama. hunch-rover-nav answers JSON dodge in a few seconds.

Sources

NASA Planetary Fact Sheet, Moon, nssdc.gsfc.nasa.gov. Heiken, Vaniman, and French, Lunar Sourcebook, Cambridge, 1991, chapter 3. Mitchell, Carrier, Houston, Scott, et al., Apollo soil mechanics experiment S-200. NASA, An Engineering Guide to Lunar Geotechnical Properties, 2022. Bekker parameters as tabulated for lunar soil in rover terramechanics references (ϕ about 31–33°, c about 170 Pa, $k_c = 1400$ N/m², k_ϕ about 8.2×10^4 N/m³). HLS-UG-001 lunar thermal analysis guidebook for the 95–390 K surface range and the one-meter thermal skin.

Code references are line-level facts from [viewer.html](#): `G` and `MU`, `gAlong`, `maxTraction`, `lunarSink`, the 0.09 m speed radius, `torque × 2`, `updateDust`'s 0.3 m/s², the 240 Wh battery, the 1.8 s model interval, and `hunch-rover-nav` as the hardcoded model.